SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

CO1 - ASSESSMENT TOOL - 1

DESCRIPTIVE WRITTEN TEST

Course Code	**CSA3717**	**Course Title**	**Software Testing**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 1: Descriptive Written Test**
Weightage	**50% of CIA**	**Total Marks**	**100**
CO1 Statement	*Apply software testing principles to enhance the effectiveness of software quality assurance and improve defect detection strategies.* **(BL3)**		
Student Name	**B Reddy Balu**	**Reg. No.**	**192312499**
Section/Batch	**Slot A / 2023**	**Date**	**21-10-2026**

Code of Conduct:

I ***B Reddy Balu (192312499),*** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Balu

Question

5. A company maintains a defect repository but fails to utilize it effectively, resulting in repeated defects across releases. As a tester, apply defect classification and repository concepts to improve the situation. Explain how defects originate and how they can be categorized. Illustrate how repository data can support test design and prevention. Suggest improvements to avoid recurring defects. Conclude with expected quality improvements.

Defect Management in Software Testing.

Introduction

In Software development, defects are inevitable. A defect repository is main-tained to store information about defects found during different testing cycles & releases.

How Defects originate

- The major Sources of defects are :-
  1. Requirements phase.
  2. Design phase.
  3. Coding phase.
  4. Integration phase.
  5. Testing phase.
  6. Deployment phase.
  7. Maintenance phase.

Defect classification

- It helps in understanding the nature, impact and root cause of defects.
- proper classification makes it easier to analyze patterns and take corrective actions.

Classification based on Severity

Severity Level	Description
Critical	Causes System crash or complete failure.
Major	Affects major functionality and no workaround available.
Minor	Affects non-critical funtionality
Trivial	Cosmetic issues, typos or minor UI issues.

Classification based on priority

High - Fix required immediately.
Medium - Fix required in next release.
Low - can be fixed in Later releases.

Defect Repository and its Importance

It is a centralized data base or tool (such as Jira, Bugzilla, HP, ALM, etc.) where all defect information is stored.
* Typically contains details such as ID.

Key benefits of maintaining an effective defect repository

- provides historical data of defects.
- Helps in identifying recurring defects.
- Supports root cause analysis.
- Guides test design and test coverage.

Suggestions for Improvement to Avoid Recurring Defects.

1. Effect use of Defect Repository.
2. Root cause Analysis. (RCA)
3. Review and Static Analysis.
4. Knowledge Sharing.
5. Improved Test Coverage.
6. Automation of R. Tests.
7. Better Req. Management.
8. Continuous Monitoring and Metrics

Conclusion:

A defect repository is a valuable asset for any organization. when defects are properly classified and utilized.

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	20%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	